

facility, or simply moving to a new geographic locale would definitely help. Of course the underground theory would depend on not just the mass and velocity of the asteroid, but also its composition. An ice-made asteroid wouldn't penetrate as deep as an asteroid made of, let's say metal, travelling at the same velocities. Newton came up with this nifty formula to calculate how deep a projectile would go if it hit something. The depth will be given by the length of the projectile multiplied by the density of the projectile divided by the density of the material its hitting ($D=LA/B$, where A = density of projectile and B = density of object being penetrated).

Why it may not happen

One word. Science! We have enough technology to spot any object in space years before it's within destructive distance of the atmosphere, and we are also sitting on enough weaponry to blow that thing to smithereens. We wouldn't worry too much about this one. Asteroids and meteors have not yet mastered the art of sneaking up on planets.

Besides, we're sitting on a nuclear arsenal large enough to blow up our planet several times over, so why couldn't we blast a rogue asteroid out of the sky? Obviously there is the concern that it might just split into still dangerous parts and hit earth in multiple places, but that's exactly why we have scientists who can come up with ways to safely knock these giant rocks out of our planet's way. There is also the possibly that burying ourselves deep into the recess of the mountain ranges like the Himalayas and the Appalachians would allow us to survive the blast. However, long term survival would only be contingent upon how well we've managed to carve out the interiors to support human life that is, fitted with air filtration systems, farms for consistently growing a food supply etc. It sounds very difficult to achieve, but is definitely doable.

Just to get you thinking, currently NASA's Jet Propulsion Laboratory is tracking close to 20 asteroids that have a potential of crashing into the earth. The good news, however is that neither of them are "planet killers" and none of them are going to be rendezvousing with us anytime in the next hundred years.



Nuclear War

World peace has become the punch line for fancy speeches to win hearts and votes. While few individuals still strive tirelessly for a nuclear weapon free planet, nations perpetuate fear amongst its citizens, justifying the ever increasing stockpile of nuclear arsenal. We're talking not just atom bombs, but hydrogen bombs too.

Currently, there's a stockpile of about 20,000 nuclear warheads, sufficient to wipe out the planet several times over. Surface to air detonation would cause a radiation fallout that would span centuries, not to mention, nuclear winter. Water would be polluted beyond drinkable levels and the land would turn toxic. Maybe now we can all have mutant super-powers. No but seriously, all it takes for this scenario to occur is one wrong move on the part of someone very nervous (or trigger happy, or egotistical or whatever).

How to survive

Surviving a nuclear war can be pretty tricky, almost bordering on impossible. Even if you do survive the initial bombing session, there's always the radiation fallout to worry about. Soon, we'd be seeing mutations in humans, and no, nothing like the Teenage Mutant Ninja Turtles. Think more like "The Hills Have Eyes." Even if we did figure out a way to survive the initial blasts, and keep away from radiation, there's still the issue of the holocaust. See, once the nuclear bombs detonate, the shockwave from the explosions are going to rock the earth's upper layer loose (no pun intended), decimating most of it into fine particulate matter. This matter, would then mix with the air and rise up into our atmosphere, where it's going to stay put. Now that there is solid rock particulate matter in the air, it's going to essentially block out the sun for years, causing temperatures to drop faster than you can say "bazzinga" and before we know it, its freezing cold and we've all gone back to the ice-age, but

only with a lot of toxic radiation to boot. Plant and vegetation life would cease to exist, so we're pretty sure the vegetarians will be the first to perish.

You may hole yourself up in a cave with plenty of food (make sure it's dehydrated) with access to a water source that originates from deep under the ground. You could even shack up in the Mountains using the same rule of supplies. However, you might not last too long because of various factors. Firstly, there is the concern of living in an enclosed space for year on end. Chances are, unless you're a Zen master, your psyche would break down. Then there is also the issue that a nuclear blast of this magnitude would decimate the Ozone layer, allowing the very harmful UV rays to reach the Earth's surface. So even if you manage to step out for a bit after the nuclear radiation has died out, chances are you're going to get toasted (quite literally) by the harmful UV rays.

Why it may not happen

Okay let's face it. Nuclear war isn't a joke and the people who hold the key to launching a nuclear strike know that. We have seen a lot of tension in modern times, including wars and despite nuclear arsenal being present, its use has not been considered an option so far. If it hasn't happened yet, it's a good chance it won't happen in the near future, but of course, that's only an optimistic assumption.

However, there is one danger. Currently, all militaries have detection systems which sound an alert when a nuclear warhead is detected as launched. The General Chief (or whoever is the top gun in the military) will only have minutes to gauge whether the warning is genuine, followed by letting the Prime Minister/ President/King know of the situation.

Further to that, the ruling authority will have fewer minutes to decide whether to retaliate or not. Now this detection system is electronic, and as with any electronic devices, they can malfunction. In fact, in 1983, Stanislav Petrov, the Lieutenant Colonel of the Soviet Air Defence Forces at the time, refrained from telling his superiors that his nuclear warhead detection system was signalling a battalion of nuclear weapons heading towards Russia. If he had done otherwise, Russia would have retaliated, launching